VOIDSCOPE · USER REQUIREMENTS

VoidScope user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade): sixteen user requirements drawn from the users and stakeholders in VDS-PRB-001; the old requirements R1 to R12 now sit in the product requirements as PR-001 to PR-013

VoidScope user requirements

What the crews, medical leads and trainers who would use VoidScope need, written from their side. The product requirements that meet these needs are in VDS-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Fits the hole we can make	The probe goes through a standard 50 mm (2 in) core hole or gap without enlarging it	High	Municipal fire and rescue crews (PRB Table 1)
UR-002	Performance	Reaches into the void	The probe reaches at least 4 m along a straight path into a void	High	Municipal fire and rescue crews; USAR trainers
UR-003	Performance	Sees in darkness and dust	The operator can read a letter or recognise a person in a dark, dusty void on the monitor	High	Municipal fire and rescue crews; volunteer teams
UR-004	Environment and durability	Survives rubble	The head and rod survive handling, drops, dust, mud and water on a collapse site	High	Volunteer and community emergency response teams
UR-005	Safety	Water cannot harm the survivor	Water reaches a survivor only in small, controlled amounts that the survivor takes by choice, under the medical lead's direction	High	Medical lead on a rescue team
UR-006	Safety	Air is clean and gentle	Air reaches the void clean and at a pressure that cannot hurt a survivor or disturb the debris	High	Medical lead on a rescue team
UR-007	Environment and durability	Works a whole shift	The kit runs a full shift on battery in cold and in heat, with no mains power	High	Volunteer and community emergency response teams
UR-008	Cost	Affordable for a municipal budget	The full kit costs a small fraction of a commercial search camera	High	Municipal fire and rescue crews; volunteer teams
UR-009	Deployment and assembly	Quick to set up and carry	Two trained people carry it and are searching within minutes of arrival	Medium	Municipal fire and rescue crews
UR-010	Maintenance	Water path stays clean	The parts that touch water are cheap and quick to replace between survivors	Medium	Medical lead on a rescue team
UR-011	Performance	Looks round a corner	The operator can look down a passage that turns away from a straight hole	High	Municipal fire and rescue crews (Amish, decision 3A, 2026-10-03)
UR-012	Usability	Talks with the survivor	The operator can speak with and hear a survivor at the tip	Medium	Medical lead on a rescue team
UR-013	Safety	Nothing hot or hazardous at the tip	The part that goes near a person is not hot enough to burn, and the battery is not a hazard	Medium	Medical lead on a rescue team
UR-014	Maintenance	Repairable with commodity parts	A team can repair the kit with common stock, hand tools and a basic 3D printer	Medium	Volunteer and community emergency response teams
UR-015	Usability	Good for drills		Low	USAR trainers

ID	Category	Title	Need	Priority	Source
			Trainers can use the same probe repeatedly in drills with a written procedure		
UR-016	Regulatory	Honest about its status	The design is published as an open engineering reference, not as certified rescue or medical equipment	Low	Stakeholders (PRB Table 2)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.